POOJA PATIL

patilpoojap16@gmail.com | (704) 652-2985 | www.linkedin.com/in/poojappatil

EDUCATION

Columbia University Mailman School of Public Health – New York, NY *May 2027*
Master of Public Health, Department of Biostatistics
Certificate in Comparative Effectiveness Outcomes Research (CEOR)

- GPA: 3.81/4.0
- *Relevant Course Work:* Research Methods and Application, Determinants of Health, Analysis of Categorical Variables, Statistical Computing Using SAS, Applied Regression
- *Involvement:* Columbia Mailman Consulting Club, Columbia Health Technology Club, Columbia Biostatistics Ambassador, Columbia 2026 Class Speaker Selection Committee

University of North Carolina at Chapel Hill – Chapel Hill, NC *May 2025*
Bachelor of Science, Quantitative Biology, Cum Laude Graduate
Bachelor of Science, Statistics and Analytics, Cum Laude Graduate
Minor in Neuroscience

- GPA: 3.54/4.0
- *Relevant Course Work:* Quantitative Personalized Genomics Lab, Stochastic Modeling, Healthcare Actuarial Risk Analytics, Optimization for Machine Learning and Neural Networks, Computational Neuroscience, Molecular Biology and Genetics, Optimization, Probability, Methods of Data Analysis

EXPERIENCE

Eli Lilly and Company – Remote *October 2026 – Present*
Clinical Project Statistician Co-Op (Global Statistical Science)

- Design statistical analysis methods and author protocol and statistical analysis plan (SAP) sections for global clinical trials, conducting analyses in SAS and R to support data quality assurance and regulatory reporting alongside physicians and clinical research scientists
- Partner with statisticians, statistical programmers, and data engineers to build and validate AI-driven automation tools across clinical development workflows, setting risk-based trust and validation standards for both generative and quality-checking AI model outputs

Varosync – New York, NY *August 2026 – October 2026*
Scientific Engagement and Program Strategy Lead

- Served as CEO's primary liaison for institutional partnership development at a Harvard, MIT, and NYC EDC-recognized AI-driven computational drug discovery startup, independently authoring strategic proposals, scoping briefs, and evidence synthesis frameworks for engagements with global pharmaceutical companies, venture capital and private equity firms, and international health foundations
- Translated a physics-informed geometric deep learning platform for GPCR drug-failure prediction into scientific and commercial narratives

Extern, PwC – Remote *June 2026 – August 2026*
PwC AI-Powered Nonprofit Consulting and Data Analytics Extern, completed through Extern in partnership with PwC
Supervisors: Juan Landi, LaRae Snoddy

- Engineered a Python-based analytical pipeline to process 25+ years of longitudinal juvenile justice data across 5 datasets, selecting and executing appropriate statistical methods to identify predictors of youth safety outcomes including youth/staff fear rates and recidivism
- Translated quantitative findings into written analytical briefs and data-driven presentations for nonprofit leadership and state agency stakeholders, applying epistemic rigor in interpreting effect size, directionality, and model limitations across all reported findings

Extern, Pfizer – Remote *May 2026 – July 2026*
Pfizer Advanced AI-Powered Document Intelligence Extern, completed through Extern in partnership with Pfizer
Supervisors: Scott Sears, Russell Vassallo, Colette McMahon

- Built end-to-end AI pipelines to process pharmaceutical vendor files using OCR and RAG frameworks, benchmarking three OCR engines across extraction accuracy, retrieval quality, and routing performance using the LlamaIndex RAG framework with open-source LLMs
- Delivered a technical evaluation report and interactive demo UI quantifying model trade-offs and optimization strategies across document intelligence system components, with systematic embedding model comparisons across Mistral-7B, MiniLM, E5, and BGE

Breast Cancer Hub – Concord, NC *March 2018 – May 2026*
Published Scientific Intern / Research Project Assistant
Supervisor: Dr. Lopamudra Das Roy, MS, MBA, PhD

- Co-authored published oncology research analyzing post-mastectomy reconstruction outcomes and developed a validated liver cancer (FLL-HCC) detection model using data synthesis and risk modeling, adopted by medical schools in India to standardize diagnostic pathways
- Co-founded a virtual fundraising initiative during COVID-19, tracking donor engagement metrics and raising $2,000+ over three months to support underserved patients and community-based cancer research programs across the United States and India

Carolina Institute of Developmental Disabilities – Chapel Hill, NC *July 2023 – May 2025*
Undergraduate Clinical Research Assistant
Supervisors: Caisi Hecht, CRC Study Coordinator, The Shen Lab/ CIDD

- Built automated data tracking systems across three neurodevelopmental protocols using REDCap, yPrime, and Excel; developed structured queries and validation checks to identify discrepancies in EEG, behavioral, and outcome datasets in a regulated research environment
- Collaborated with research coordinators and clinical teams to streamline data collection workflows, conducting root-cause analyses on dataset discrepancies to reduce turnaround time between clinical assessments and database entry across all three protocols

Internet Tobacco Vendors Study (ITVS) – Chapel Hill, NC *Nov 2022 – April 2023*
Remote Undergraduate Task Research Assistant
Supervisor: Jessie Scott, Internet Tobacco Vendors Study

- Collected, cleaned, and structured longitudinal datasets to evaluate federal regulatory compliance among online tobacco vendors, applying data wrangling techniques to prepare large-scale data for regulatory enforcement analysis
- Conducted quantitative and qualitative analyses of digital marketing and sales practices, identifying statistically meaningful patterns in nicotine product promotion across vendor platforms

LEADERSHIP

Columbia Mailman School of Public Health, Biostatistics Department – New York, NY *May 2026 – Present*
Quantitative Specialist + Graduate Teaching Assistant (Integration of Science and Practice) + Biostatistics Student Ambassador

- Provide individualized biostatistics coaching to MPH students across quantitatively heavy academic subjects within research methods and applications, including applied regression, categorical data analysis, and statistical computing in SAS and R
- Facilitate classroom seminar discussion, reinforce applied public health methodology across core MPH curricula, and guide incoming students in connecting foundational epidemiological theory to real-world public health decision-making and practice
- Lead peer engagement panels, represent the department at school-wide events, and bridge communication between incoming students and Biostatistics faculty leadership

Columbia Mailman Consulting Club – New York, NY *March 2026 – Present*
Vice President of Communications
- Manage communications between the club, consulting firms, and industry partners to coordinate events and company engagements; develop data-driven findings through healthcare strategy case consulting and structure analytical insights

GoPeer – Concord, NC *10+ Years*
Independent Private Tutor
- Designed data-informed lesson plans and tracked performance longitudinally across 100+ K-12 and college-level students, applying trend analysis and outcome measurement to drive a 34% average improvement in test scores overall

CERTIFICATIONS

Forage, Oliver Wyman Integrated Consulting Group Job Simulation Certificate *June 2026*
- Analyzed financial and operational data for an insurance client, diagnosed underperformance across key metrics, and developed client-facing deliverables including a hypothesis deck, KPI dashboard, and executive strategy presentation

Forage, PwC US Management Consulting Job Simulation Certificate *June 2026*
- Conducted transaction diligence for a simulated M&A engagement, built a financial model with editable assumptions in Excel, and synthesized synergy potential into an executive presentation with a stakeholder summary for J&J

SKILLS & TECHNICAL COMPETENCIES

- **Programming & Statistical Software:** R, Python (pandas, NumPy, scikit-learn), SAS, STATA, MATLAB, REDCap, yPrime, Excel, Tableau; familiarity with Power BI
- **Statistical Methods:** Regression Analysis (Linear, Logistic, Lasso), Random Forest, Gradient Boosting (GBM), Predictive Modeling, Survival Analysis, Categorical Data Analysis, Stochastic Modeling, Simulation, Optimization, Feature Engineering
- **AI & Machine Learning:** RAG Pipeline Development, LlamaIndex, LLM Benchmarking and Evaluation, OCR Evaluation (Tesseract, PaddleOCR, EasyOCR), HuggingFace Embeddings, Mistral-7B, TinyLlama, Phi-2, Prompt Engineering
- **Research & Clinical:** Statistical Analysis Plan (SAP) Authoring, Clinical Trial Coordination, Data Wrangling, Quantitative and Qualitative Analysis, Scientific Writing, Root-Cause Analysis, Data Validation and Integrity, Health Informatics, Public Health Analytics, Real-World Evidence, Outcomes Research, ICH Guidelines, GCP Principles, CDISC Standards (foundational)
- **Consulting & Strategy:** Hypothesis-Based Problem Solving, Financial Modeling, Market Access Analysis, Client-Facing Deliverable Development, Executive Presentation, Stakeholder Communication, Healthcare Strategy, Commercial Analytics, Institutional Partnership Development
- **Software & Tools:** Microsoft Office Suite (Word, Excel, PowerPoint, Outlook), Google Suite, LlamaIndex, GitHub; familiarity with AWS SageMaker
- **Languages:** English (native), Hindi (conversational), Marathi (conversational)
- **Interests and Involvement:** Indian Classical Dance Graduate (12+ Years), GoPeer Tutor (10+ Years), Recreational Soccer (5 Years)

PROJECTS

AI-Powered Pharmaceutical Document Intelligence Pipeline | *Pfizer x Extern Externship*
- Benchmarked chunking strategies and hybrid retrieval configurations across the LlamaIndex RAG framework, evaluating trade-offs between dense and sparse retrieval, reranking approaches, and multi-model LLM routing logic; assessed context window utilization and hallucination rates across Mistral-7B, TinyLlama, and Phi-2 to produce a deployment recommendation grounded in domain-specific pharmaceutical document retrieval requirements

Longitudinal Juvenile Justice Predictive Analytics Pipeline | *PwC x Extern Externship*

- Matched research questions to appropriate statistical methods across five longitudinal datasets spanning facility climate, incident reports, administrative records, staff surveys, and outcome measures; applied Python-based data cleaning and merging across heterogeneous data structures with documented decisions around missingness handling, variable selection, and interpretation of effect size relative to policy-relevant thresholds

Undiagnosed Glycemic Conditions in U.S. Adults | *NHANES Population-Level Analysis*

- Applied nested logistic regression in SAS to NHANES survey data, selecting Youden's J index for optimal classification threshold determination and Cochran-Mantel-Haenszel testing to evaluate associations stratified by age, BMI, and race/ethnicity; performed sensitivity analyses incorporating survey weighting and complex sampling design to produce population-representative estimates of undiagnosed diabetes and prediabetes prevalence

HN Adam Optimizer Implementation | *Undergraduate Machine Learning Research*

- Implemented the HN Adam adaptive optimization algorithm from scratch in PyTorch, benchmarking convergence speed, loss trajectory, and final test accuracy against standard Adam and SGD across neural network training tasks; analyzed sensitivity to learning rate schedules and batch size configurations, contributing to methodological documentation of adaptive gradient descent variants in deep learning optimization

Predictive Modeling of NBA Game Outcomes | *Sports Analytics Research*

- Engineered game-level and player-level features across 25,000+ observations spanning 20 NBA seasons in R, comparing Lasso regression, Random Forest, GBM, and stepwise regression on holdout test sets; evaluated bias-variance trade-offs across models and selected optimal configuration based on out-of-sample predictive accuracy and variable importance rankings for interpretability in a sports performance context